COMP 4736
Introduction to Operating Systems

School	School of Computing and Academic Studies
Program	Computer Systems Technology (CST) Diploma
Course Credits	3
Minimum Passing Grade	50%
Start Date	January 05, 2026
End Date	April 24, 2026
Total Hours*	45
Total Weeks	15
Hours/Weeks	3
Delivery Type	Lecture
Prerequisite(s)	• 50% in COMP 2510 and 50% in COMP 2721
CRN	90341

Acknowledgement of Territories

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the Coast Salish Nations of Sḵwx̱wú7mesh [1] (Squamish), səl̓ilwətaʔɬ [2] (Tsleil-Waututh), and xwməθkwəy̓əm [3] (Musqueam).

Instructor Details

Name	Rahim Oraji
E-mail	roraji@bcit.ca
Location	Online
Office Hours	By appointment via video conferencing

Course Description

This course introduces basic concepts in operating systems with a focus on using concurrent programming: processes and threads, inter-process communication (IPC), synchronization and mutual exclusion, deadlock and starvation with an introduction to memory organization. Supporting theory includes overall operating system structure, operating systems services, how these services are used, and kernel architecture. To illustrate the concepts, each topic includes examples of using these services in programs using an operating system. With the concepts learnt in this course, the students will be able to analyze existing concurrent solutions and design and implement concurrent software applications.

Course Learning Outcomes/Competencies

Upon successful completion of this course, the student will be able to:

- Describe the role of an operating system.
- Analyze how the operating system is structured (layers and components).
- Describe the operating system interface (system calls, shells).
- Use POSIX system calls in C.
- Describe synchronization concepts such as mutual exclusion, critical sections, and race conditions.
- Solve synchronization problems using constructs such as spinlocks, mutexes, semaphores, condition variables, and monitors.
- Compare and contrast synchronization of threads with semaphores and monitors.
- Discuss the consequences of concurrency: deadlock, starvation and livelock.
- Describe how to prevent, avoid, detect and recover from deadlock.
- Design and implement concurrent applications using processes and kernel-level and user-level threads.
- Solve IPC problems by implementing applications using methods such as signaling, messages, pipes, and shared memory.

Learning Resources

1. Modern Operating Systems (5th Edition) by Andrew Tanenbaum. (**Recommended**)
2. Operating Systems: Three Easy Pieces,Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau (University of Wisconsin-Madison). Online version [4] (**Further reading -- Optional)**
3. Operating Systems – Internals and Design Principles, William Stallings, Ninth Edition. (**Further reading -- Optional**)

Evaluation Criteria

Criteria	%	Comments
Final exam	35	
Miterm	30	
Lab assignments	30	There is a weekly lab assignment which must be completed by each student individually during lab time.
Participation	5	

Attendance Requirements

Regular attendance in lectures, seminars, and labs is seen as critical to student success, and may be monitored by faculty.

Unapproved absence of 2 or more classes may result in failure or forced withdrawal from the course or program.

Please see Policy 5101 - Student Regulations: https://www.bcit.ca/files/pdf/policies/5101.pdf [5]

Other information

Important dates :

1. First day of classes: Jan 5
2. Midterm exam : Feb 23 - Feb 27
3. Computing reading week: Mar 16 - Mar 20
4. Last day of classes Apr 10
5. Final exam : Apr 20 to Apr 24

Course Schedule and Assignments

Week	Date	Material Covered	Reference Chapter (Tanenbaum)	Marked Work
1	**Jan 6**	Computer architecture and operating system overview 1	Chapter 1 and lecture notes	
2	**Jan 13**	Computer architecture and operating system overview 2	Chapter 1 and lecture notes	

3	**Jan 20**	Operating system structures and clocks	Chapter 1 and 5
4	**Jan 27**	Introduction to processes Process model, creation, termination, states, implementation, modeling multiprogramming, and Linux process states	Chapter 2 and lecture notes
5	**Feb 3**	Introduction to threads Thread usage, the classical thread model, POSIX threads, implementing threads in user space and Kernel, hybrid Implementations, and pop-Up threads	Chapter 2
6	**Feb 10**	Process management, system calls	Chapter 1, 2, and lecture notes
7	**Feb 17**	Concurrency 1 Race conditions, critical regions, and mutual exclusion with busy waiting (disabling interrupts, lock variables, strict alternation, Peterson's solution, the TSL instruction)	Chapter 2
8	**Feb 23 -27**	Midterm	No classes
9	**Mar 3**	Concurrency 2 Sleep and wakeup, semaphores, mutexes, monitors, message passing, and barriers	Chapter 2
10	**Mar 10**	Concurrency 3 and Deadlocks 1 IPC in Linux, system V IPC vs POSIX IPC, pipes, named pipes, shared memory, message queues, semaphores, signals, and memory mapping Preemptable and nonpreemptable resources, resource acquisition, conditions for resource deadlocks, deadlock modeling, Ostrich algorithm,	Chapter 2, chapter 6, and lecture notes
	Mar 16 - 20	Computing Reading Week	
11	**Mar 24**	Deadlocks 2 Deadlock detection and recovery,deadlock avoidance, banker's algorithm, deadlock prevention, livelock, and starvation	Chapter 6
12	**Mar 31**	Introduction to memory management 1 No memory abstraction vs memory abstraction, base and limit registers, swapping, memory management with bitmap and linklist, and memory management algorithems..	Chapter 3
13	**Apr 7**	Introduction to memory management 2 Virtual memory, paging, page fault, page tables, translation lookaside buffers(TLB), page tables for large memories, and two level page table and inverted page table.	Chapter 3
14	**Apr 14**	Final exam review	

15	**Apr 20 - 24**	Final Exam

*** This is a tentative schedule and subject to change ***.

BCIT Policy

The following statements are in accordance with the BCIT Policies 5101, 4501, 5103, 5104, and 7507, and their accompanying procedures. To review these policies and procedures please click on the links below.

Attendance
In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with their instructor or Program/Department Head, stating the reason for the absence. When absences result in missed safety requirements, exams, or other deadlines, an instructor or Program/Department Head may request appropriate supporting documentation, including a medical note. For all other absences, the student is responsible to ensure they seek out missed information, preferably from a classmate or recording (when available). Please see BCIT Policy 5101 - Student Regulations [6] , and accompanying procedures, for more information.

Attempts
As stated in BCIT Policy 5103 - Student Evaluation [7] , students must successfully complete a course within a maximum of three (3) attempts at the course. Students with two attempts in a single course will be allowed to repeat the course only upon permission from the Associate Dean. Students who have not successfully completed a course within three attempts will not be eligible to graduate from their respective program. For those courses or programs that have Education Council approval, the number of attempts as stated in the evaluation section of the course outline shall apply.

Academic Integrity
It is the responsibility of all students to be familiar with the Student Code of Academic Integrity. Violations of the Code, including plagiarism, cheating, misrepresentation, and academic advantage, are prohibited and will be handled in accordance with BCIT Policy 5104 – Student Code of Academic Integrity [8] , and accompanying procedures.

Accommodation
Any student who may require accommodation from BCIT because of a physical or mental disability should refer to BCIT's Policy 4501 - Accommodation for Students with Disabilities [9] , and contact BCIT's Accessibility Services (SW1 2360, 604-451-6963) at the earliest possible time. Requests for accommodation must be made to Accessibility Services, and should not be made to a course instructor or Program area.

Any student who needs special assistance in the event of a medical emergency or building evacuation (either because of a disability or for any other reason) should promptly inform their course instructor(s) and Accessibility Services of their personal circumstances.

Human Rights, Harassment and Discrimination
The BCIT community is made up of individuals from every ability, background, experience and identity, each contributing uniquely to the richness and diversity of the BCIT community as a whole. In recognition of this, and the intrinsic value of our diversity, BCIT seeks to foster a climate of collaboration, understanding and mutual respect between all members of the community and ensure an inclusive accessible working and learning environment where everyone can succeed. Respect, Diversity, and Inclusion [10] is a supportive resource for both students and employees of BCIT, to foster a respectful learning and working environment. Any student who feels that they are experiencing discrimination or harassment (personal or human rights-related) can confidentially access this resource for advice and support. Please see BCIT Policy 7507 – Harassment and Discrimination [11] and accompanying procedure.

Students should make themselves aware of additional Education, Administration, Safety and other BCIT policies listed at https://www.bcit.ca/about/administration/policies.shtml [12]

Guidelines for School of Computing and Academic Studies

No school specific policies, please refer to main BCIT Policy.

Approved

I verify that the content of this course outline is current.
Rahim Oraji, Instructor
December 30, 2025

I verify that this course outline has been reviewed.
Chi En Huang, Program Head
January 02, 2026

I verify that this course outline has been reviewed and complies with BCIT policy.
Donna Turner, Associate Dean
January 05, 2026

Note: Students will be given reasonable notice if changes are required to the content of this course outline.

*Course hours and credits are calculated per Policy 5012 [13] and the associated procedure [14] .

Total hours – Example of 3 credit lecture/lab course:

- ***Full-time course****: 45 hours of scheduled learning*
- ***Flexible Learning course****: 36 hours of scheduled learning plus 9 hours of independent (non-scheduled, non-instructional) learning*

List of links found on this page

This list includes all links found on this page for your reference.

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[2] https://twnation.ca/
[3] https://www.musqueam.bc.ca/
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